

PRINCIPAL COMPONENT ANALYSIS AND FACTOR ANALYSIS FOR STANDARDIZED AND UNSTANDARDIZED DATA SETS

Data Sets: FOODP.DAT, MASST.DAT and
AUDIO.DAT

Principal Component Analysis on food.csv

DATA: FOODP.DAT

FOODP.DAT provides average prices of hamburgers, apples, bread, tomatoes, and butter in cents per pounds in 24 US Cities in 1978. The data is collected by the US Department of Labor, Bureau of Labor Statistics (Washington DC,1978). This data set is relevant because it shows the different average prices from five common food items within the same time frame, conditions, and quality. From an economic standpoint, high prices for these items would induce a higher consumer purchasing power. Based on customer purchasing power, the cost of living can be deduced in these metropolitan cities since the cost-of-living index is correlated to customer purchasing power.

Our clustering analysis would be unstandardized for this data sets because prices have the same unit (cents) and similar values within a reasonable margin of error. So, the original scale would remain the same. The goal is to cluster cities based on the singular effect on prices for these five variables. Initially, single linkage clustering would be useful for this dataset because we only have 24 cities and 5 variables. It should be noted that consumer purchasing power is a relative measure that needs to be analyze over time due to inflation. Therefore, a clustering analysis for the same products, recorded in the same conditions year after year would give a more detailed analysis of the evolution of cost of living and consumer purchasing power of clustered metropolitan US cities over time.

Step 1: Load the data

- Process is the same for both standardized and unstandardized data sets.

Step 2: Define the problem in terms of principal components

- Point of references is x_1 (bread), x_2 (hamburger), x_3 (butter), apples(x_4), tomato (x_5),(x_6),(x_7), (x_8). We matched the eigenvectors in step 3 to Eq1. To Eq. 5 (Appendix, PCA).
- Point of references are the same for both standardized and unstandardized data.

Step 3: Show the value of principal components

- **Food(U)**
- Without standardizing the data, we have y1 is more important than all other components combined (Table 1). All eigenvectors associated with y1 are negative. Therefore, low y1 would be correlated to the most expensive city and vice versa.
- Also e[2,1] which represent hamburger has the highest absolute value(magnitude) for the first principal component. So, a high price of hamburger and bread are highly associated with a low value of principal component 1. Butter and hamburgers are highly associated with a low value of principal component y2. And apples are highly associated with a high value of y3.
- Interpretation: Anchorage has the lowest y1 therefore hamburger and bread are expensive there in 1978. Kansas City has the lowest y2 therefore butter and hamburgers were expensive there. Both facts are probably due to transportation reasons. Finally, Anchorage has high value y3 therefore apples were expensive there due to the weather as a hypothesis.

```
> eigenvalues
[1] 216.79440 79.12794 62.26846 34.67047 22.22005
> eigenvectors
      [,1]      [,2]      [,3]      [,4]      [,5]
[1,] -0.4529089 0.05515147 0.21435116 0.6856702 0.52511130
[2,] -0.7146773 0.48679539 -0.02261341 -0.1338116 -0.48358009
[3,] -0.3391656 -0.75632931 -0.43256354 0.1976187 -0.29456459
[4,] -0.2203644 -0.42895099 0.81242257 -0.3231370 -0.05470465
[5,] -0.3471543 -0.06289354 -0.32619100 -0.6070257 0.63296724
```

Figure 1A: eigenvalues and eigenvectors for unstandardized food data set

- **Food (S)**
- Bread and hamburgers are still important for y1 however apples are the most important for y2(opposite to butter and hamburgers in the unstandardized data set). And butter is highly associated with a low value y3(opposite to apples in the unstandardized data sets).

```
> eigenvalues
[1] 2.4393555 0.9295911 0.8332378 0.5328728 0.2649429
> eigenvectors
      [,1]      [,2]      [,3]      [,4]      [,5]
[1,] -0.5099267 -0.05649609 0.4017162 0.53197875 0.54074549
[2,] -0.5200718 0.27761601 0.4074371 -0.07148075 -0.69378685
[3,] -0.3973106 -0.09940133 -0.7684773 0.43041438 -0.23759156
[4,] -0.2909422 -0.87675286 0.0713631 -0.37257819 -0.05243928
[5,] -0.4764421 0.37571444 -0.2774329 -0.62275040 0.40872301
```

Figure 1B: eigenvalues and eigenvectors for standardized Food data set

Step 4: Check variance estimates of PCs vs original X and other properties

- When the data set is standardized, each food item has equal importance going into principal components which is not the case for the unstandardized data sets (**Figure 2**).

```
> var1+var2+var3+var4+var5
[1] 5
> percentvars
  Bread Hamburger Butter Apples Tomato
    0.2         0.2    0.2     0.2    0.2
>
> var1+var2+var3+var4+var5
[1] 415.0813
> percentvars
  Bread Hamburger Butter Apples Tomato
0.1686394 0.3260335 0.2051058 0.1683354 0.1318860
```

Figure 2: Variance and percent variance for standardized and unstandardized data set

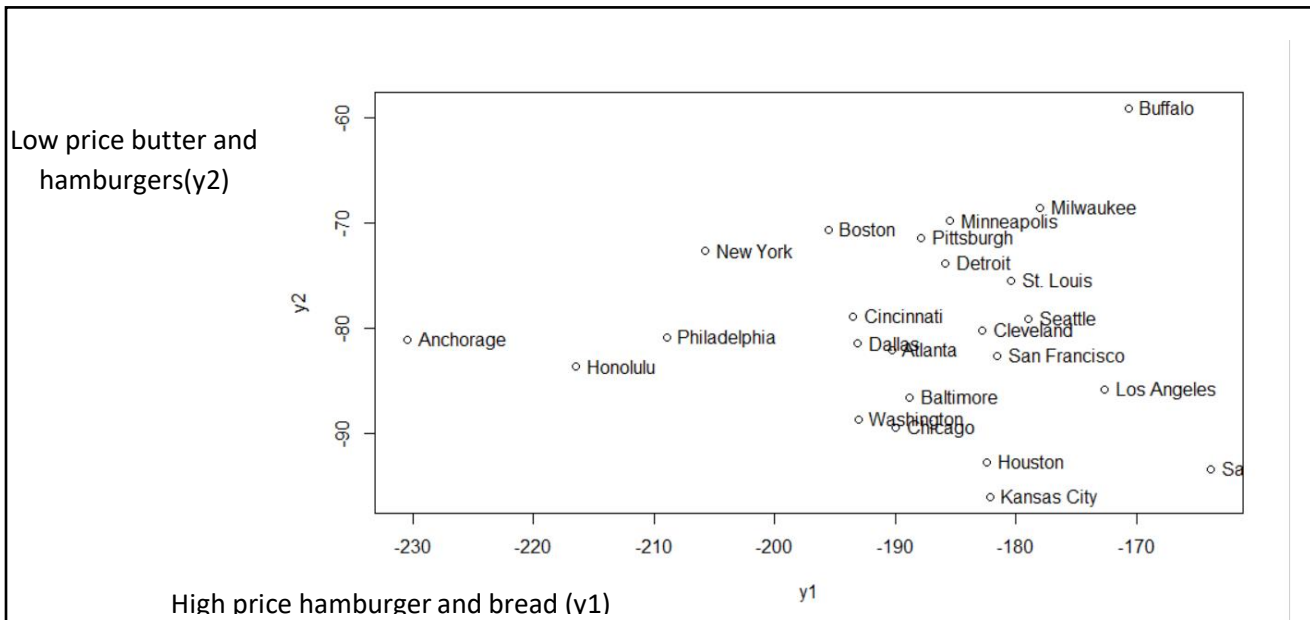
Total Variance Table	eigenvalues (U)	%(U)	eigenvalues (S)	%(S)
y1	216.7944	0.5222938	2.439	0.488093
y2	79.12794	0.1906324	0.929	0.185912
y3	62.26846	0.1500151	0.833	0.1667
y4	34.67047	0.0835269	0.532	0.106464
y5	22.22005	0.0535318	0.264	0.052832
Total	415.08132	1	4.997	1

Table 1: Total variance Table for Food.csv(U). Based on that table, y1, y2 and y3 are the most relevant principal components. This is also verified with standardized data based on the disparity between eigenvalues.

- Finally, they are uncorrelated regardless of if the data sets is standardized or not.

Step 5: Visual Plot and interpretation

- Both plots (standardized vs unstandardized) are similar. Certain cities had noticeable shift such as Buffalo, but most cities remained in the same area plot.



Step 6: Use them in a regression

- Both standardized and unstandardized data sets have the same R square. However, we get different R square if we use principal components in the regression. It makes sense because by standardizing the data (setting up the variance equals for all food items) , we have restructure the principal components.

```
> summary(lm(dv~as.matrix(y1+y2)))

Call:
lm(formula = dv ~ as.matrix(y1 + y2))

Residuals:
    Min       1Q   Median       3Q      Max
-34.791 -11.440   0.328   5.245  27.622

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   -12.9798    50.5846  -0.257    0.8
as.matrix(y1 + y2)  -1.6774     0.1875  -8.946 8.79e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15.47 on 22 degrees of freedom
Multiple R-squared:  0.7844,    Adjusted R-squared:  0.7746
F-statistic: 80.03 on 1 and 22 DF, p-value: 8.793e-09

>
```

```

> summary(lm(dv~y1+y2))

Call:
lm(formula = dv ~ y1 + y2)

Residuals:
    Min       1Q   Median       3Q      Max
-17.887  -5.892  -1.040   6.604  16.593

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  13.7498    17.9181   0.767  0.4514
y1           -2.4511     1.2660  -1.936  0.0664 .
y2             0.1431     0.2223   0.644  0.5266
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.375 on 21 degrees of freedom
Multiple R-squared:  0.1558,    Adjusted R-squared:  0.0754
F-statistic: 1.938 on 2 and 21 DF,  p-value: 0.1689

```

Figure 3: Regression summary highlighting the change in principal component structure between standardized and unstandardized data set.

Step 7 Standardize variables

- After standardizing the data, we see that Anchorage and Honolulu are expensive cities. Because the z scores for all five food items are positive. For instance, we have 2.00 standard deviation above the mean for hamburger at Anchorage. Standardized data are in line with unstandardized analysis from that standpoint. To conclude, we did see a change in principal components from standardized to unstandardized data. However, the analysis was similar overall (as shown in the plot) because all food items have similar price within a reasonable margin of error.

```

> data=scale(data)
> data=as.data.frame(data)
> data

```

	Bread	Hamburger	Butter	Apples	Tomato
Anchorage	3.879538966	2.0075514	1.16909152	1.45501855	1.39773432
Atlanta	-0.244027483	-0.0641127	0.00943799	0.25870320	0.83008074
Baltimore	-1.140454972	-0.2962078	0.73557618	-0.50693863	1.99241903
Boston	0.568733440	0.6063844	-0.23983333	-1.27258046	0.91117411
Buffalo	-0.471122447	-0.2016506	-2.10394929	-1.93055390	-1.87303157
Chicago	-0.160360917	-0.4079574	0.12865471	1.59857639	0.60031619
Cincinnati	-0.160360917	0.5032310	0.58384581	-0.73423855	0.14078710
Cleveland	0.006972214	-0.3907651	-0.16396815	-0.17197033	-0.88639558
Dallas	-0.351598782	0.3914815	-0.18564391	1.27557125	0.12727154
Detroit	0.281876644	-0.2962078	-0.44575312	-1.44006461	0.85711186
Honolulu	1.489065662	1.6723028	1.10406422	1.58661324	0.55976951
Houston	-0.399408248	-0.8549554	0.65971100	0.90471349	-0.71069328
Kansas City	-0.399408248	-1.0698583	1.96025702	-1.09313315	-0.25116418
Los Angeles	-0.184265650	-1.3793185	-0.41323947	0.35440842	-1.41350247
Milwaukee	-0.614550845	-0.2704195	-2.27735542	0.71330303	-0.27819531
Minneapolis	-0.710169777	0.3828854	-0.98764729	-0.44712286	-0.08897745
New York	0.508971608	1.5949378	0.48630486	-0.49497548	0.31648940
Philadelphia	0.532876341	1.2596892	1.03903692	0.01944012	1.58695218
Pittsburgh	-0.184265650	0.2711358	-0.57580772	-0.94957531	0.28945828
St. Louis	-0.184265650	-0.2102467	-0.45659100	-0.60264386	-1.45404916
San Diego	-0.710169777	-2.3850641	0.18284413	-0.38730710	-1.00803563
San Francisco	0.186257712	-0.6572447	-0.55413195	0.89275033	-1.06209788
Seattle	-0.746026877	-0.5884758	-0.80340327	0.27066635	-0.15655525
washington	-0.793836343	0.3828854	1.14849954	0.70133988	-0.42686648

Figure 4: Standardized data output for Food File

Principal Component Analysis on audio.csv

DATA: AUDIO.DAT

AUDIO.DAT gives audiometric measures for 100 males of 39 years old in 1991. Each subject was exposed to a signal with specific frequency that was incrementally increased until the signal was perceived. This was done by a calibrated audiometer. The unit of the device is in decibel. However, all measurement is in decibel loss. It represents the difference between the base value from the device and the decibel value at which the signal was perceived. Two observations are made because a signal might be perceived by one ear before the other for any subject. Furthermore, observations were obtained for frequencies 500Hz,1000Hz,2000Hz and 4000Hz therefore there are 8 data points for each subject. Finally, the range is from -10 to 99, some values are at -10 but the maximum value never reached the maximum value in the range. We will discuss our decision on standardizing the data since the dataset is in the same unit.

Procedure and Interpretation

Step 1: Explore the data

Process:

- We set 8 equalities in our input from var 1 to var 8 based on the different frequencies. The data could then be interpreted after being initially pull into R.

```
> rho=cor(data)
> rho
```

	L500	L1000	L2000	L4000	R500	R1000	R2000	R4000
L500	1.0000000	0.7775422	0.4012198	0.2553577	0.6962869	0.6416175	0.2371881	0.2040887
L1000	0.7775422	1.0000000	0.5365505	0.2749451	0.5515407	0.7070264	0.3597445	0.2168873
L2000	0.4012198	0.5365505	1.0000000	0.4250181	0.2391181	0.4459829	0.7011440	0.3262145
L4000	0.2553577	0.2749451	0.4250181	1.0000000	0.1789801	0.2631959	0.3164523	0.7097405
R500	0.6962869	0.5515407	0.2391181	0.1789801	1.0000000	0.6634387	0.1588895	0.1321084
R1000	0.6416175	0.7070264	0.4459829	0.2631959	0.6634387	1.0000000	0.4142321	0.2201095
R2000	0.2371881	0.3597445	0.7011440	0.3164523	0.1588895	0.4142321	1.0000000	0.3745589
R4000	0.2040887	0.2168873	0.3262145	0.7097405	0.1321084	0.2201095	0.3745589	1.0000000

```
> cov(data)
```

	L500	L1000	L2000	L4000	R500	R1000	R2000	R4000
L500	41.07071	37.72727	28.13131	32.10101	31.78788	26.30303	14.12121	25.28283
L1000	37.72727	57.32323	44.44444	40.83333	29.74747	34.24242	25.30303	31.74242
L2000	28.13131	44.44444	119.69697	91.21212	18.63636	31.21212	71.26263	68.98990
L4000	32.10101	40.83333	91.21212	384.77525	25.01010	33.02525	57.66667	269.11869
R500	31.78788	29.74747	18.63636	25.01010	50.74747	30.23232	10.51515	18.19192
R1000	26.30303	34.24242	31.21212	33.02525	30.23232	40.91919	24.61616	27.21717
R2000	14.12121	25.30303	71.26263	57.66667	10.51515	24.61616	86.30303	67.26263
R4000	25.28283	31.74242	68.98990	269.11869	18.19192	27.21717	67.26263	373.66414

```
> summary(data)
```

	L500	L1000	L2000	L4000	R500	R1000	R2000
Min.	:-10.0	Min. :-10.0	Min. :-10.00	Min. :-10.00	Min. :-10.0	Min. :-10.0	Min. :-10.0
1st Qu.	:-10.0	1st Qu. :-5.0	1st Qu. :-5.00	1st Qu. : 8.75	1st Qu. :-10.0	1st Qu. :-5.0	1st Qu. :-5.0
Median	:-5.0	Median : 0.0	Median : 0.00	Median : 20.00	Median : -5.0	Median : 0.0	Median : 0.0
Mean	:-2.8	Mean : -0.5	Mean : 2.00	Mean : 21.35	Mean : -2.6	Mean : -0.7	Mean : 1.6
3rd Qu.	: 5.0	3rd Qu. : 5.0	3rd Qu. : 6.25	3rd Qu. : 35.00	3rd Qu. : 0.0	3rd Qu. : 5.0	3rd Qu. : 5.0
Max.	: 15.0	Max. : 20.0	Max. : 45.00	Max. : 70.00	Max. : 25.0	Max. : 20.0	Max. : 35.0

```
R4000
Min. :-10.00
1st Qu.: 5.00
Median : 15.00
Mean : 21.35
3rd Qu.: 30.00
Max. : 75.00
```

Figure 5: Correlation, covariances matrices and data summary

Interpretation:

- Based on the correlation matrix (**Fig.5**), there is a connection between each ear for the same subject. Indeed, the observation at a specific frequency from one ear is close to the one on the other. Since all subjects are 39, there is not a disparity between each ear which might happen for elderly people. Furthermore, we will standardize the data because the variance is dominated by the top frequencies. By doing so, all frequencies would be weighted equally on principal components. This is counter intuitive because the data set is in the same units. Therefore, having a data set with the same units does not mean that the data should be automatically unstandardized.

Step 2: Defining the problem in term of principal components

Process:

- We input `data=scale(data)` before the vars equalities to standardize the data in R

Interpretation:

- As discussed above, the weight of higher frequencies is overwhelming. Therefore, we define the problem with an equal percent weight at 12.5%.

```
> percentvars
      L500      L1000      L2000      L4000      R500      R1000      R2000      R4000
0.03557445 0.04965200 0.10367862 0.33328303 0.04395624 0.03544322 0.07475360 0.32365885

> percentvars
      L500 L1000 L2000 L4000      R500 R1000 R2000 R4000
0.125 0.125 0.125 0.125 0.125 0.125 0.125 0.125
.
```

Figure 6: Percent contribution to total variance for both unstandardized and standardized data

Step 3: Compute all eigenvalues and eigenvectors in R

Process:

- Eigenvalue and eigenvectors are calculated in R via sigma. Indeed `eigenvalues=eigen(sigma)$values` while `eigenvectors=eigen(sigma)$vectors`.

```
> eigenvalues
[1] 3.9290053 1.6183218 0.9753248 0.4667822 0.3400900 0.3158912 0.2001111 0.1544736

> eigenvectors
      [,1]      [,2]      [,3]      [,4]      [,5]      [,6]      [,7]      [,8]
[1,] -0.4010948 0.3169638 -0.15815686 -0.3277576 -0.02313643 0.44590406 -0.32925525 0.54629986
[2,] -0.4209908 0.2254640 0.05196134 -0.4816310 0.37922678 -0.06745818 0.03312121 -0.62273890
[3,] -0.3663748 -0.2385933 0.47029298 -0.2824293 -0.43924664 -0.06379987 0.52551666 0.18634686
[4,] -0.2808559 -0.4741545 -0.42950248 -0.1610807 -0.35031958 -0.41692698 -0.42694396 -0.08393472
[5,] -0.3432510 0.3860197 -0.25931925 0.4876003 -0.49750307 0.19477711 0.15935065 -0.34253018
[6,] -0.4114209 0.2317725 0.02885395 0.3723164 0.35131760 -0.61363774 0.08367787 0.36136545
[7,] -0.3115483 -0.3170590 0.56293305 0.3914171 0.11078565 0.26503010 -0.47781584 -0.14658750
[8,] -0.2542212 -0.5135121 -0.42622285 0.1590982 0.39595899 0.36604656 0.41393534 0.05082062
```

Figure 7: Eigenvalues and eigenvectors

Interpretation

- Eigenvalues gives the total variance with respect to principal component. We have 8 principal components. Principal components 1 has a value of 3.929 while the last component 8 is at 0.1544. In other word principal component is 25 times bigger than the last component. Right away we know that we must drop that last component in our final model. This approach would justify dropping components 4 to components 7. For the first three components, there is a significant change in slope from component 2 to component 3 as well. Therefore, I would choose either the first 2 components or the first 3 components and the first components as a last option. Now if we look at eigenvectors, we that all values are negative for principal component 1(**Fig.7**). Therefore, they are all going in the same direction as they aggregate the variance value for PC1. At PC2, negative values are associated with high frequencies and positive values are correlated to low frequencies. Opposite to PC1 which is correlated to the sensitivity of the subjects,

PC2 is sensitive to high vs low frequencies. PC3 is more difficult to analyze. I would initially take a model with PC1 and PC2. However, PC4 have negative values for all L frequencies and positive values for R frequencies. This is an important layer in our analysis therefore PC4 should be included. Therefore, final model includes PC1 to PC4.

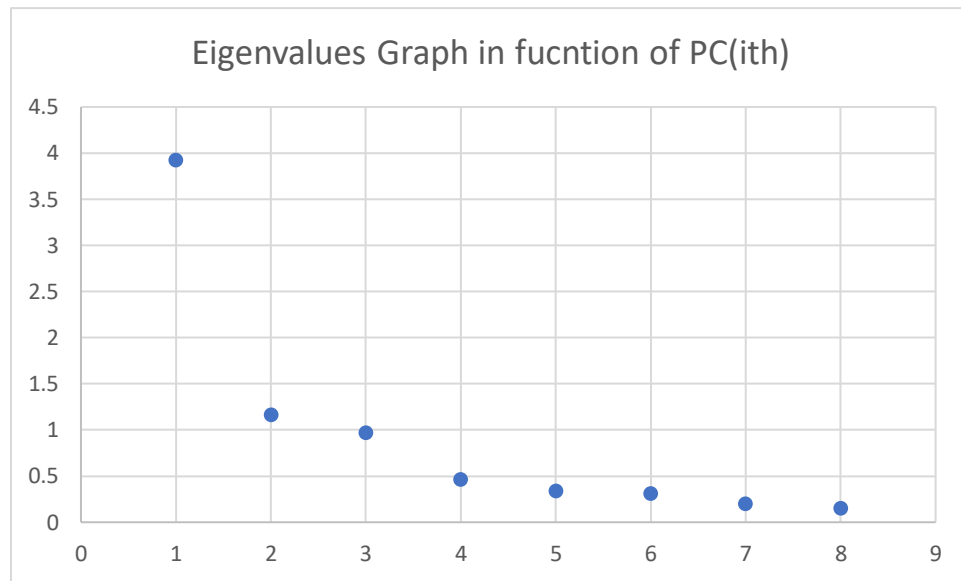


Figure 8: Eigenvalues Plot

[Step 4: Check variance estimates of the principal components and all other properties](#)

```
> percentvars
L500 L1000 L2000 L4000 R500 R1000 R2000 R4000
0.125 0.125 0.125 0.125 0.125 0.125 0.125 0.125
> percentvars_pc=eigenvalues/sum(eigenvalues)
> percentvars_pc
[1] 0.49112566 0.20229022 0.12191560 0.05834777 0.04251125 0.03948640 0.02501389 0.01930921
```

Figure 9: From percentvars to percentvars_pc

Interpretation

- Since the data was standardized, we had similar value for each component at 0.125. However, when we look at percentvars_pc, PC1 is much higher than the other components at 0.49. Then, PC2 is at 0.20 which is much lower than PC1 but also much higher than the other components. From percentvars value, we can see that our model should not include more than the first three principal components indeed.

```
> ## D: Magnitude of eigenvectors are importance of kth variable in the ith PC
> eigenvectors
      [,1]      [,2]      [,3]      [,4]      [,5]      [,6]      [,7]      [,8]
[1,] -0.4010948  0.3169638 -0.15815686 -0.3277576 -0.02313643  0.44590406 -0.32925525  0.54629986
[2,] -0.4209908  0.2254640  0.05196134 -0.4816310  0.37922678 -0.06745818  0.03312121 -0.62273890
[3,] -0.3663748 -0.2385933  0.47029298 -0.2824293 -0.43924664 -0.06379987  0.52551666  0.18634686
[4,] -0.2808559 -0.4741545 -0.42950248 -0.1610807 -0.35031958 -0.41692698 -0.42694396 -0.08393472
[5,] -0.3432510  0.3860197 -0.25931925  0.4876003 -0.49750307  0.19477711  0.15935065 -0.34253018
[6,] -0.4114209  0.2317725  0.02885395  0.3723164  0.35131760 -0.61363774  0.08367787  0.36136545
[7,] -0.3115483 -0.3170590  0.56293305  0.3914171  0.11078565  0.26503010 -0.47781584 -0.14658750
[8,] -0.2542212 -0.5135121 -0.42622285  0.1590982  0.39595899  0.36604656  0.41393534  0.05082062
```

Figure 10: Magnitude of eigenvectors.

Step 5: Visual plots for interpretations

Process

- Plot function is used in R to show any graph. Also, xlab (x axis) and ylab (yaxis) can be determined as it shown below.

Interpretation

- From the control charts which represents y_1 vs y_2 , y_1 vs y_3 respectively (Fig.11), we can see that the plots are less homogenous as you increase in principal component with respect to y_1 . There are indeed less outliers when you plot y_1 vs y_2 than y_1 vs y_3 .
- Percent variance vs ith vector plot gives the limit of necessary variable in the model. Indeed, PC variance line crosses with our original line at PC3. Therefore, only the three first components are required to optimize variance in the analysis.

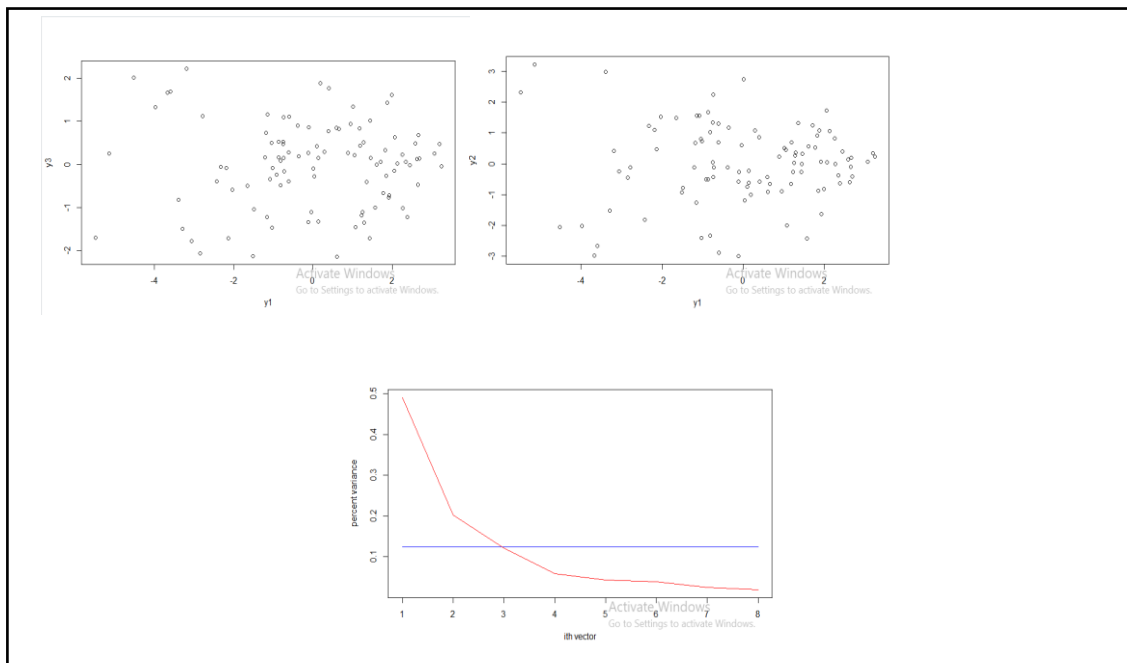


Figure 11: Percent variance vs PC; PC1 vs PC2 and PC1 vs PC3

Step 6: Use PCs as input for regression

Process

- In R, we ran a linear regression with the first two principal components and then we complete the same process with the first three components. The objective is to determine how similar a model with only 2 and 3 variables is in line with the original dataset with all 8 variables.

```
> summary(lm(dv~y1+y2))
Call:
lm(formula = dv ~ y1 + y2)

Residuals:
    Min       1Q   Median       3Q      Max
-18.516  -5.661  -2.423   3.984  18.308

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.3610     0.9032   2.614  0.0104 *
y1          -2.9441     0.4580  -6.428 4.84e-09 ***
y2          -0.5273     0.7136  -0.739  0.4617
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.032 on 97 degrees of freedom
Multiple R-squared:  0.3015,    Adjusted R-squared:  0.2871
F-statistic: 20.94 on 2 and 97 DF,  p-value: 2.766e-08

> summary(lm(dv~y1+y2+y3))
Call:
lm(formula = dv ~ y1 + y2 + y3)

Residuals:
    Min       1Q   Median       3Q      Max
-19.871  -5.288  -1.711   4.195  18.697

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.3610     0.8954   2.637  0.00976 **
y1          -2.9441     0.4540  -6.485 3.84e-09 ***
y2          -0.5273     0.7074  -0.745  0.45780
y3           1.5009     0.9112   1.647  0.10280
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.954 on 96 degrees of freedom
Multiple R-squared:  0.3207,    Adjusted R-squared:  0.2995
F-statistic: 15.11 on 3 and 96 DF,  p-value: 3.957e-08
> |

> summary(lm(dv~as.matrix(y)))
Call:
lm(formula = dv ~ as.matrix(y))

Residuals:
    Min       1Q   Median       3Q      Max
-20.249  -5.235  -1.348   4.341  20.153

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.3610     0.8927   2.645  0.00963 **
as.matrix(y)1 -2.9441     0.4526  -6.504 4.14e-09 ***
as.matrix(y)2 -0.5273     0.7053  -0.748  0.45657
as.matrix(y)3  1.5009     0.9085   1.652  0.10197
as.matrix(y)4 -0.6897     1.3132  -0.525  0.60073
as.matrix(y)5  2.3139     1.5385   1.504  0.13605
as.matrix(y)6  1.7281     1.5963   1.083  0.28187
as.matrix(y)7 -2.0819     2.0057  -1.038  0.30203
as.matrix(y)8  2.0231     2.2828   0.886  0.37783
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.927 on 91 degrees of freedom
Multiple R-squared:  0.3599,    Adjusted R-squared:  0.3036
F-statistic: 6.396 on 8 and 91 DF,  p-value: 1.429e-06
```

Figure 12: linear regression summary

Interpretation

- Based on the R2 values, we can see that the minimum number of principal components that fits the model best is three. It is counter intuitive because our eigenvectors show that PC4 had relevant correlation based on the type of audiometric devices (L vs R), Therefore the model does not affect by the type of audiometric devices but only its frequencies.

Factor Analysis on audio.csv

Steps: Explore the data (1); compute eigenvalues and eigenvectors of the correlation matrix (2); compute the estimated factor loadings (3); compute the common variance and unique variance (4); Check the model to reproduce correlation (5); create residual matrix (6); Plot pair of loadings to interpret factor loadings (7) and reproduce loading plots with rotation (8)

Process & Interpretation:

- Factor 1 is loading the 4000Hz frequency.
- Factor 2 is loading 500 Hz, 1000 Hz, & 2000 Hz frequencies.
- We decide to keep the data unstandardized. The drawback is that factors with the highest variations will compound on the first factor. However, this is acceptable because 4000 Hz frequency is associated the most patients with hearing loss.

Factors	Eigenvalue greater than 1	Eigenvalue
1	TRUE	3.92
2	TRUE	1.62
3	FALSE	0.9753
4	FALSE	0.4668
5	FALSE	0.3401
6	FALSE	0.3159
7	FALSE	0.2
8	FALSE	0.1545

Figure 13: Eigenvalues-correlation matrix. We see that only the first two factors have an eigenvalue superior to 1. These two factors represent almost 70% of the total variance. Hence common factors is equal to 2 ([step 1-2](#)).

	[,1]	[,2]
[1,]	-0.7950389	0.4032200
[2,]	-0.8344760	0.2868202
[3,]	-0.7262179	-0.3035224
[4,]	-0.5567047	-0.6031874
[5,]	-0.6803825	0.4910683
[6,]	-0.8155068	0.2948454
[7,]	-0.6175423	-0.4033411
[8,]	-0.5039101	-0.6532555

Figure 14: Factor Loading Estimation Matrix. As it is the case for Principal component 1, factor 1 has the most relationship with all variables. As you switch to other factors, they represent less proportion of the total variance ([Step 3](#)).

	round.common..4.	round.unique..4.
1	0.7947	0.2053
2	0.7786	0.2214
3	0.6195	0.3805
4	0.6738	0.3262
5	0.7041	0.2959
6	0.7520	0.2480
7	0.5440	0.4560
8	0.6807	0.3193

Figure 15: Reproduce correlation Table. The factor 1 has a similar value than the one in the original correlation matrix. However, there is a difference for other factors especially factor 7 and factor 8. It can be hypothesized that there is an aggregation of error as you go from one factor to the others (Step 4-5).

	L500	L1000	L2000	L4000	R500	R1000	R2000	R4000
L500	0.00000000	-0.001550302	-0.05376538	0.05597307	-0.04265220	-0.125629760	-0.09114683	0.066866221
L1000	-0.001550302	0.00000000	0.01759538	-0.01660535	-0.15707055	-0.058062118	-0.03989333	-0.016246749
L2000	-0.053765384	0.017595384	0.00000000	-0.16235175	-0.10593770	-0.056760605	0.13025072	-0.238011723
L4000	0.055973067	-0.016605354	-0.16235175	0.00000000	0.09641409	-0.012953620	-0.27062667	0.035175834
R500	-0.042652199	-0.157070549	-0.10593770	0.09641409	0.00000000	-0.036207185	-0.06320742	0.110049879
R1000	-0.125629760	-0.058062118	-0.05676061	-0.01295362	-0.03620718	0.00000000	0.02954547	0.001776749
R2000	-0.091146832	-0.039893327	0.13025072	-0.27062667	-0.06320742	0.029545466	0.00000000	-0.200111684
R4000	0.066866221	-0.016246749	-0.23801172	0.03517583	0.11004988	0.001776749	-0.20011168	0.00000000

Figure 16: Residual Matrix: The higher the residuals the more difference reproduced correlation is to our correlation matrix. Again, it is evident that Factor 1 and factor 2 are the only necessary variable for our model (Step 6).

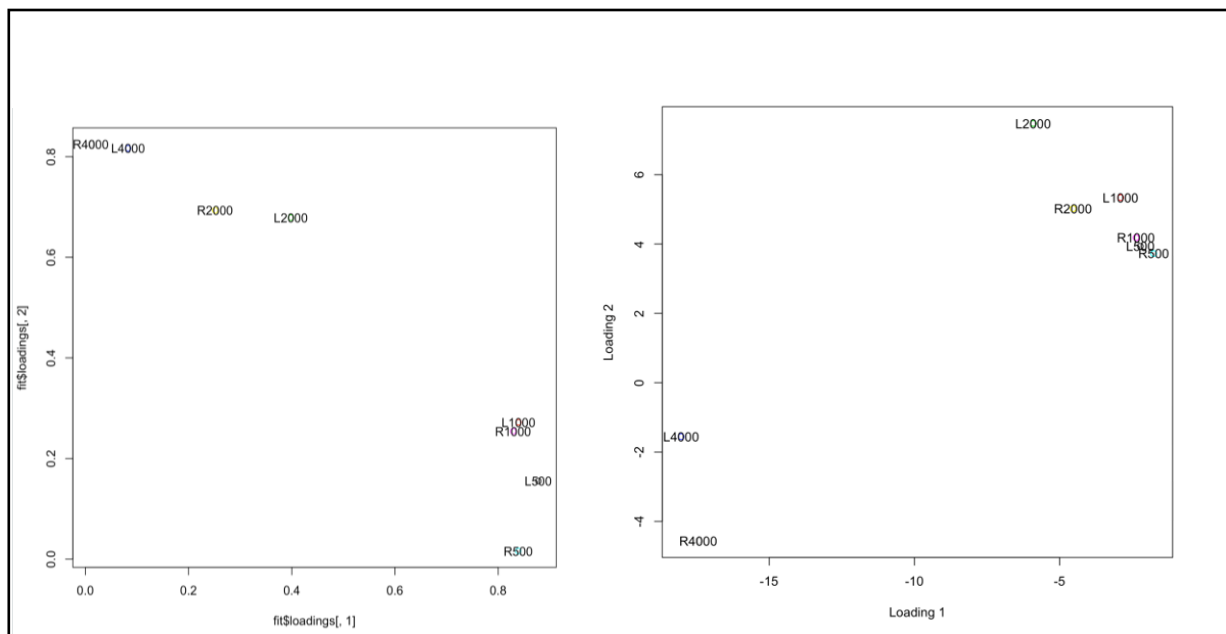


Figure 17: Loading Plots. Without rotation, it was impossible to interpret the data. However as shown in Fig.17, Factor 1 is loading 500Hz and 1000Hz which are low to medium decibel loss.

However, Factor 2 is loading 2000Hz and 4000Hz according to our plot. The result is in line with principal component analysis of the same data set. ([Step 7-8](#)).

[Factor Analysis on masst.csv](#)

MASST.DAT provides the result of a survey to estimate the potential demand for mass transportation. As oil prices are volatile, there was a sharp increase in oil prices during the Arab Oil Embargo in 1973-1974. Therefore, this survey was conducted in three southern cities to study the impact on this political decision on mass transportation demand. The survey is based on the demographics of the partakers and their views on mass transportation as oil prices have increased. The data set has 52 variables described in **Table 12**. There are two main differences between this data set and FOODP.DAT. First, there are over 1000 missing data therefore mean substitution would be used to analyze the data set. Furthermore, even though variables do not have different units, and all are following a numerical method of rankings; all variables should be on the same scale. Therefore, we are going to standardize the data. In other words, R would modify the variables to a mean of 0 and a standard deviation of 1. We want to compare the relevance of each variable with each other in the dataset. Single linkage clustering would be inappropriate due to the dimension of the data.

[Steps: Explore the data \(1\); compute eigenvalues and eigenvectors of the correlation matrix \(2\); compute the estimated factor loadings \(3\); compute the common variance and unique variance \(4\); Check the model to reproduce correlation \(5\); create residual matrix \(6\); Plot pair of loadings to interpret factor loadings \(7\) and Reproduce loading plots with rotation \(8\)](#)

Process & Interpretation:

- **Missing Data:** We learn two techniques to deal with missing data: data omission and mean imputation. The latter technique was used to replace the empty cells in Excel. There is an error associated with this technique as it shrinks standard errors. However, the limitations of the technique would be neglected in our analysis as it is the most adequate technique. Indeed, there are too many missing data to omit missing cells and design a proper model as well.

```
> eigenvalues
[1] 8.91521252 3.76965229 3.06650435 2.66866010 2.46707197 2.42120455 1.89113213 1.72795477 1.58091335 1.39023223
[11] 1.27685783 1.17020097 1.14674960 1.02484934 1.00561943 0.98079974 0.93532816 0.87637415 0.86020522 0.80764201
[21] 0.78831785 0.74985365 0.70387106 0.67453147 0.66812591 0.63593341 0.59621127 0.58181426 0.55569657 0.52840291
[31] 0.51688704 0.50302795 0.47981953 0.45521287 0.42794951 0.40609352 0.38637737 0.36856890 0.36009771 0.33059460
[41] 0.30452148 0.27087702 0.24544268 0.13342682 0.07522653 0.06636763 0.05213043 0.04402596 0.03828763 0.03439853
[51] 0.02027745 0.01446577
> (eigenvalues)>1
[1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE FALSE FALSE FALSE FALSE
[20] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[39] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
```

Figure 18: Eigenvalues-Correlation Matrix. We see that only the first 15 factors have an eigenvalue superior to 1. These two factors represent almost 70% of the total variance. Hence common factors are initially equal to 15 (step 1-2). Contrary to FA **AUDIO.DAT**, we computed eigenvalue >1 in R. Furthermore variables 10 through 15 would be dropped in our model because they have one variable with a loading. Therefore, common factor is equal to 9 ([Step 1-2](#))

We have listed below the summary of the 9 factors with both variables and explanations associated to them based on our model and information provided by **Table 2**. ([Step 3 through 8](#))

Factor 1: Variable 5 & Variables 8 through 18

- Respondent with low energy use per passenger as they commute to school or work. The economic strain is relevant if there is an increase in gas prices.

Factor 2: Variable 7,43,51,52 & Variable 46 through 48

- Respondents with high social status (education, income) therefore they have a driver license and owns a car. They do not utilize mass transit in general, but they would not use mass transit if there were an increased gas price.

Factor 3: Variable 20,21,23,27,28 & Variable 30,31,33,34,36

- Respondents with a different political view on energy than the current US administration or the rise in price does not affect them much. (V20, V21, V30, V31, V36). On the other hand, (V20, V21, V30, V31, V36) represents the respondents at the other end of this spectrum.

Factor 4: Variable 41,42 & 44

- Marital status is associated with an increased number of passengers per households. So, they are paying attention to mass transit prices fluctuations.

Factor 5: Variable 1 through 4 & Variable 6, 26, 38

- Respondents that pay attention to economic strain associated with an increased in gas prices. They would be against government pro-Israel stand if it affects their bottom line as it related to transportation.

Factor 6: Variable 22,24,35 & Variables 37,40,45

- Variable 22 represents respondents with favorable energy crisis opinions. They may be Republicans. Or simply the rise in prices does not affect their bottom line enough for them to sway their political beliefs. On the other hand, variables 24,35 &37 represents respondents with the opposite opinion.

Factor 7: Variable 32,39

- Variable 32 represents respondents with unfavorable energy crisis opinions. It should be noted that respondents might be willing to use mass transit as they have resided in the area for a long time. Therefore, they would be affected by a price hike in public transportation. Also, the higher the respondents' age, the more revenue is spent on commuting to work. In other words, there is a probable increase in energy consumption.

Factor 8: Variables 29. 39

- Variable 29 represents respondents that are less likely to utilize mass transit and/or take additional steps to reduce energy consumption. Respondents who have a long commute to school or work are less likely to utilize mass transit, thereby increasing their energy consumption.

Factor 9: Variable 19,25

- Variable 19&25 want improvement in mass transit system. It starts with affordability of public transportation. Therefore, they would indeed be impacted by an increase in gas price, but it would not affect their support of the Reagan Administration on energy.

Conclusion

PCA and FA are data reduction techniques. The objective is to create a model based on the necessary variables that is like the initial one with all variables initially associated with the data set. The input is identical as shown between **FA AUDIODAT** and **PCA AUDIO.DAT**. Also, the output look the same as it is the case for the plots. However, the plot does not show the same thing. Indeed, PCA is a linear combination of variable which is adequate for data sets such as **AUDIO.DAT** and **FOODP.DAT**. However, FA which generate a measurement model of a latent variable i.e.-a variable that is inferred and not observed- is more adequate for **MASST.DAT**.

Appendix

The following is a brief description of the data in file MASST.DAT
(Questions 4.9, 5.12, 6.9, 7.5, 8.9)

Variable	Variable Location (Columns)	Description
ID	1-4	Identification number from 1 to 598
V1	5-6	Economy
V2	7-8	Convenience
V3	9-10	Flexibility
V4	11-12	Safe from dangerous people
V5	13-14	Low energy use per passenger
V6	15-16	Dependability
V7	17	10% increase
V8	18	20% increase
V9	19	30% increase
V10	20	40% increase
V11	21	50% increase
V12	22	60% increase
V13	23	70% increase
V14	24	80% increase
V15	25	90% increase
V16	26	100% increase
V17	27	150% increase
V18	28	> 150% increase
V19-V38	29-48	Energy crisis opinions
V39	49	Miles to work or school
V40	50	Age
V41	51	Gender
V42	52	Marital Status
V43	53	Driver's License
V44	54	Number of adults in household
V45	55	Number of children in household
V46	56	Number of cars in household
V47	57	Respondent's education
V48	58	Spouse's education
V49	59	Years of residence in city
V50	60	Number of full-time workers
V51	61	Annual household income
V52	62	Ethnic background

Table 2: Description of all 52 variables for Mass Transit